



Data Converters

Chapter 03

Spectral Metrics

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VLSI Data Conversion Circuits

held by

Prof. B. Murmann, PhD, University of Hawai'i

during his sabbatical at the University of Freiburg in 2009



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Spectral Metrics – What we want to understand

- SNR – signal-to-noise-ratio
- SNDR – signal-to-noise-and-distortion ratio
- ENOB – effective number of bits
- DR – dynamic range
- SFDR – spurious free dynamic range
- THD – total harmonic distortion

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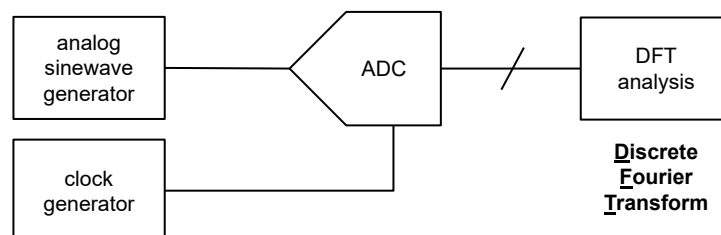
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Dynamic Performance Metrics

frequency domain

- performance metrics result from considering the output spectrum of the ADC
- basic idea: apply one or more tones at converter input
→ expect same tone(s) at output, all other frequency components represent non-idealities
- both static errors, e.g., DNL and INL, and dynamic errors contribute to frequency domain non-idealities



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Discrete Fourier Transform – Basics

- Discrete Fourier Transform (DFT) takes a vector of N time domain samples (spaced $T_s = 1/f_s$) and computes a set of N frequency bins

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi k \frac{n}{N}}$$

- bin k represents frequency content at $k * f_s / N$
- DFT frequency resolution
 - proportional to $f_s / N = 1 / T_s / N$, unit: Hz/bin
 - $N * T_s$ is total time spent on gathering samples
- A DFT with $N = 2^{\text{integer}}$ can be found using a computationally efficient algorithm: the Fast Fourier Transform (FFT)

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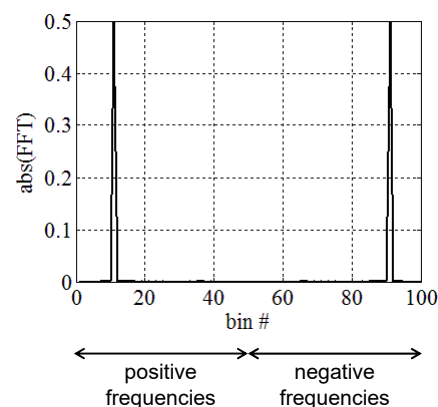
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Example – Sine Wave

```
% signal frequency
fsig = 100;
% signal amplitude
amp = 1;
% sampling frequency
fs = 1000;
% # of samples
N = 100;

% sampled sine wave
x = amp*cos(2*pi*fsig*[0:N-1]/fs);
% magnitude of FFT
xfft = abs(1/N*fft(x));

figure(1)
clf
plot(xfft, 'k', 'linewidth', 2);
xlabel('bin #');
ylabel('abs(FFT)');
```



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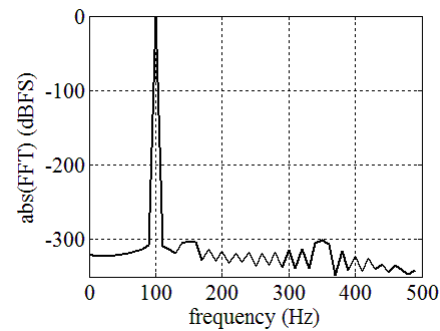


Normalized Plot with Frequency Axis

```
% signal frequency
fsig = 100;
% signal amplitude
amp = 1;
% sampling frequency
fs = 1000;
% # of samples
N = 100;

% sampled sine wave
x = amp*cos(2*pi*fsig*[0:N-1]/fs);
% magnitude of FFT
xfft = abs(1/N*fft(x));
% single-side FFT in dBFS
xfft = 20*log10(2*xfft(1:end/2)/amp);
% frequency vector
f = [0:N/2-1]*fs/N;

figure(1)
clf
plot(f,xfft, 'k', 'linewidth', 2);
xlabel('frequency (Hz)');
ylabel('abs(FFT)');
```



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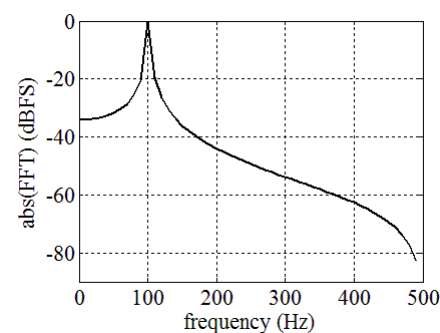
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Another Example

```
% signal frequency
fsig = 101;
% signal amplitude
amp = 1;
% sampling frequency
fs = 1000;
% # of samples
N = 100;

% sampled sine wave
x = amp*cos(2*pi*fsig*[0:N-1]/fs);
% magnitude of FFT
xfft = abs(1/N*fft(x));
% single-side FFT in dBFS
xfft = 20*log10(2*xfft(1:end/2)/amp);
% frequency vector
f = [0:N/2-1]*fs/N;

figure(1)
clf
plot(f,xfft, 'k', 'linewidth', 2);
xlabel('frequency (Hz)');
ylabel('abs(FFT)');
```



Spectrum of a sinusoid?
→ What's wrong?

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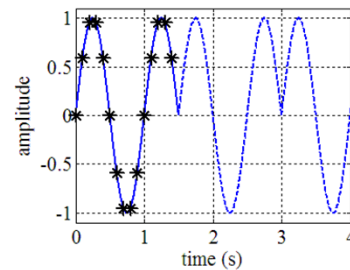
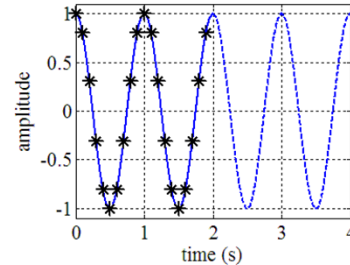
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Spectral Leakage

- DFT assumes that data repeats every N samples
- a sequence that contains a non-integer number of sine wave cycles has discontinuities in its periodic repetition
 - discontinuity looks like a high frequency signal component
 - power spreads across spectrum
- two approaches to deal with this
 - integer number of cycles
 - windowing



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Integer Number of Cycles

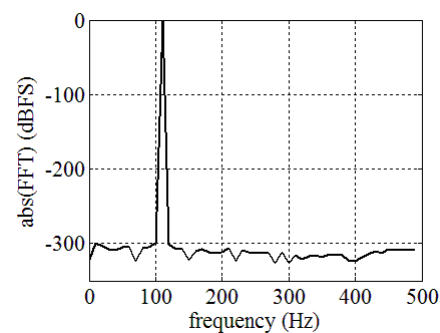
```
% signal frequency
fsig = 110;
% signal amplitude
amp = 1;

% sampling frequency
fs = 1000;
% # of samples
N = 100;
% bin width = fs/N = 10

% sampled sine wave
x = amp*cos(2*pi*fsig*[0:N-1]/fs);
% magnitude of FFT
xfft = abs(1/N*fft(x));
% single-side FFT in dBFS
xfft = 20*log10(2*xfft(1:end/2)/amp);
% frequency vector
f = [0:N/2-1]*fs/N;

figure(1)
plot(f,xfft, 'k', 'linewidth', 2);
xlabel('frequency (Hz)');
ylabel('abs(FFT)');
```

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Note: $\# \text{ periods} = T_{\text{sim}}/T_{\text{period}}$

$$= (N/f_s) / (1/f_{\text{sig}})$$

$$= \frac{100}{1000} * 110$$

$$= 11$$

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Windowing

- spectral leakage can be attenuated by windowing the time samples before performing the DFT
- windows taper smoothly down to zero at the beginning and the end of the observation window
- time domain samples are multiplied by window coefficients on a sample-by-sample basis
 - means convolution in frequency
 - sine wave tones and other spectral components smear out over several bins
- many different window functions available
 - trade-off: attenuation $\leftrightarrow$ smearing

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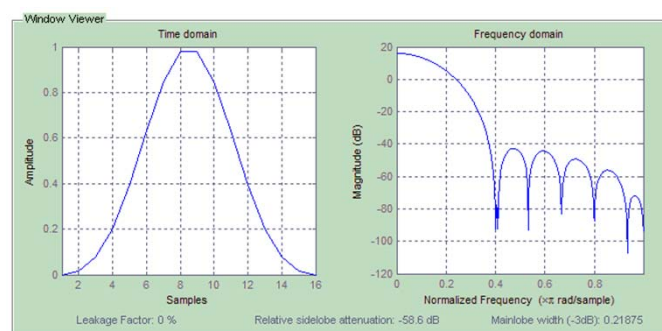
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Example – Blackman Window

```
N = 16;  
wvtool(blackman(N));
```



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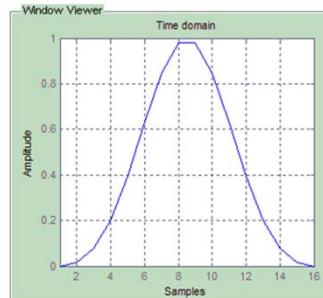
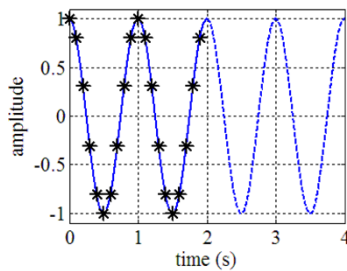
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Windowing

- spectral leakage can be attenuated by windowing the time samples before performing the DFT
- windows taper smoothly down to zero at the beginning and the end of the observation window
- time domain samples are multiplied by window coefficients on a sample-by-sample basis



`N = 16;
wvtool(blackman(N));`

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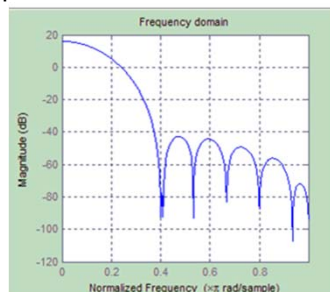
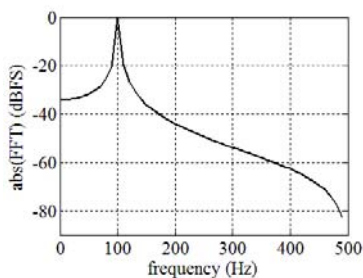
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Windowing

- time domain samples are multiplied by window coefficients on a sample-by-sample basis
 - means convolution in frequency
 - sine wave tones and other spectral components smear out over several bins



`N = 16;
wvtool(blackman(N));`

- many different window functions available
 - trade-off: attenuation ↔ smearing

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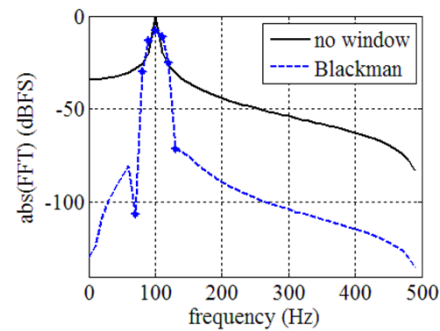


Another Example – Spectrum with Window

```
% signal frequency
fsig = 101;
% signal amplitude
amp = 1;
% sampling frequency
fs = 1000;
% # of samples
N = 100;

% sampled sine wave
x = amp*cos(2*pi*fsig/fs*[0:N-1]);
% sampled & windowed sine wave
x_windowed = amp*cos(2*pi*fsig*[0:N-1] / fs)'.*blackman(N);
% magnitude of FFT
xfft = abs(1/N*fft(x));
xfft_windowed = abs(1/N*fft(x_windowed));
% single-side FFT in dBFS
xfft = 20*log10(2*xfft(1:end/2)/amp);
xfft_windowed = 20*log10(2*xfft_windowed(1:end/2)/amp);
% frequency vector
f = [0:N/2-1]*fs/N;

figure(1)
clf
plot(f,xfft, 'k', 'linewidth', 2);
hold on
plot(f,xfft_windowed, 'b--', 'linewidth', 2);
plot(f(8:14),xfft_windowed(8:14), 'b*', 'linewidth', 2);
xlabel('frequency (Hz)');
ylabel('abs(FFT) (dBFS)');
grid on
legend('no window', 'Blackman')
```



Signal power “smears” over several bins,
i.e., it “leaks” into neighboring bins!

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Comparison - Complete Cycles vs. Windowing

- integer number of cycles
 - test signal falls into single bin
 - careful choice of signal frequency → ideal for simulations
 - in lab: lock sampling & signal frequency generator (“coherent sampling”)
- windowing
 - no restrictions on signal frequency
 - signal and harmonics “distribute” over several bins
 - requires more samples for given accuracy

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SQNR – Simulation versus Calculation

```
% sampled & quantized sine wave
x = round(amp*cos(2*pi*fsg*[0:N-1]/fs)/LSB)*LSB;
% function for calculating the PSD, also performs windowing
[Pxx, W] = pwelch(x, blackman(N), 0, N, fs);
% extract PSD without signal
Pxx_noise = [Pxx(1:bin_sig-2); Pxx(bin_sig+4:end)];
% calculate power of quantization noise
P_noise_dBFS = 10*log10(sum(Pxx_noise)/((FS/2)^2/2)*(fs/N))
P_noise_dB = 10*log10(sum(Pxx_noise)*(fs/N))
% calculate signal power
P_signal_dBFS = 10*log10(sum(Pxx(bin_sig-2+1:bin_sig+2+1))/((FS/2)^2/2)*(fs/N))
P_signal_dB = 10*log10(sum(Pxx(bin_sig-2+1:bin_sig+2+1))*(fs/N))
% calculate SQNR
SQNR_dBFS = P_signal_dBFS - P_noise_dBFS
SQNR_dB = P_signal_dB - P_noise_dB
% plot PSD
figure(1)
clf
plot(W, 10*log10(Pxx/((FS/2)^2/2)*(fs/N)), 'k', 'linewidth', 2);
hold on
% highlight signal bins
plot(W(bin_sig-2+1:bin_sig+2+1), 10*log10(Pxx(bin_sig-2+1:bin_sig+2+1)/((FS/2)^2/2)*(fs/N)), 'r');
% format plot
grid on
axis([0 fs/2 -110 0])
xlabel('frequency (Hz)');
ylabel('PSD (dBFS)');
```

```
% signal frequency outside bin
fsg = 101;
% signal amplitude
amp = 1;
% sampling frequency
fs = 1000;
% # of samples = 2^12
N = 4096;
% signal frequency on bin
bin_sig = ceil(fsg/(fs/N));
fsg = bin_sig*fs/N;
% full scale
FSR = 2;
% # bits
B = 10;
% least significant bit
LSB = FSR/2^10;
```

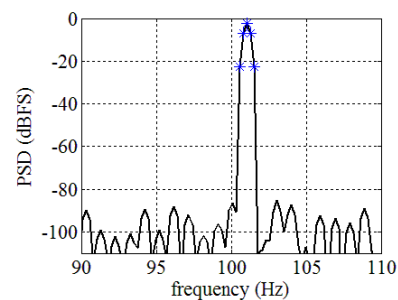
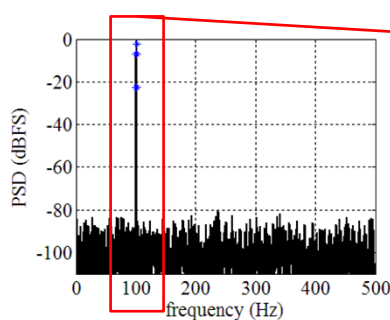
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SQNR - Spectrum of Quantized Signal



$$SQNR = \frac{\text{signal power}}{\text{quantization noise}} = \frac{1/2(FSR/2)^2}{1/12 (FSR/2^B)^2}$$

calculation: SQNR = 61.97dB

simulation: SQNR = 61.90dB

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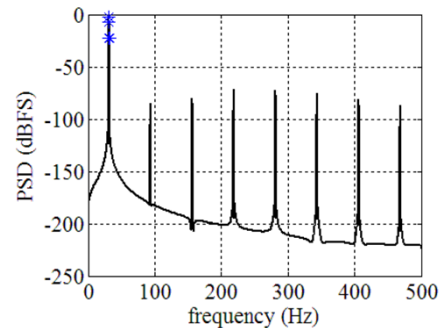
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Periodic Quantization Noise

```
% signal amplitude
amp = 1;
% sampling frequency
fs = 1000;
% # of samples = 2^12
N = 4096;
% signal frequency on
bin
fsig = 128*fs/N;
.
```

$$\frac{N}{bin\#} = 32$$



Quantization error and thus quantization noise is highly deterministic and periodic.

Make N and $bin\#$ mutually prime in order to get a more random and “white” quantization noise.

→ greatest common divisor $GCD(N, bin\#) = 1$

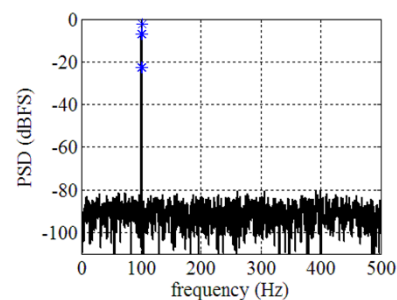
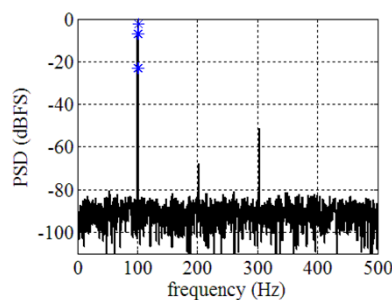
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ADC – Typical Output Spectrum & SNR



- fairly uniform noise floor due to additional electronic noise
 - harmonics due to nonlinearities, e.g., INL

- signal-to-noise ratio

$$SNR = \frac{\text{signal power}}{\text{total noise power}}$$

- total noise power includes all bins except DC, signal and harmonics

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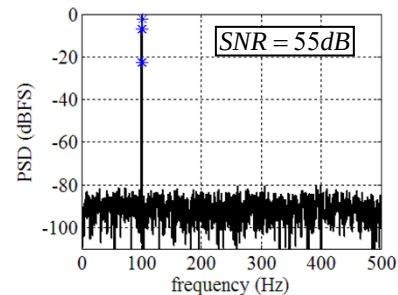
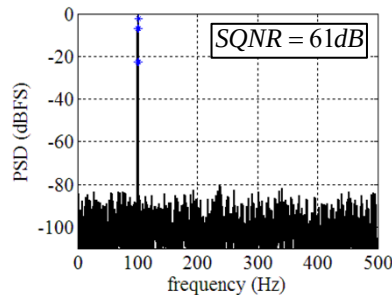
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SQNR versus SNR



- both quantization noise and electronic noise determine SNR
- typically $SNR = SQNR - 1\text{bit}$, i.e., the quantization noise is designed 1bit or 6 dB lower than the thermal noise
 - reason: cutting thermal noise down by a factor 2x
→ power dissipation increases by a factor 4x

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SNDR & ENOB

- signal-to-noise-and-distortion ratio

$$SNDR = \frac{\text{signal power}}{\text{noise \& distortion power}}$$

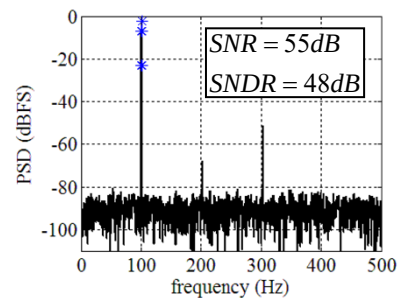
- noise & distortion power includes all bins except DC and signal

- effective number of bits

$$ENOB = \frac{SNDR(dB) - 1.76dB}{6.02dB}$$

- here: $ENOB = \frac{48dB - 1.76dB}{6.02dB} = 7.7$ ← stated: 10-bits ADC

→ ideal ENOB, i.e., SQNR, only for zero electronic noise, perfect transfer function with zero INL,...



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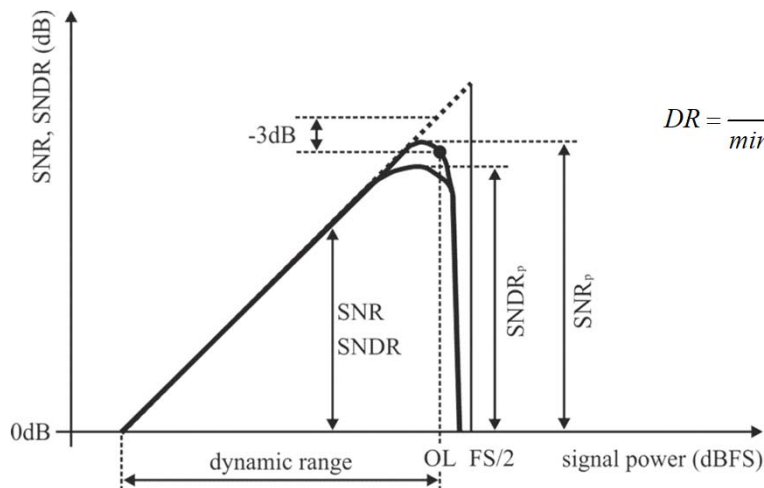
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Dynamic Range – DR



dynamic range

$$DR = \frac{\text{maximum signal power}}{\text{minimum detectable signal power}}$$

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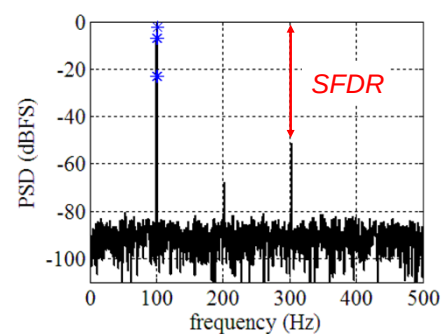
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Spurious Free Dynamic Range – SFDR

- spurious free dynamic range

$$SFDR = \frac{\text{signal power}}{\text{largest spurious power}}$$

- largest spur is often (but not necessarily) a harmonic of the input tone



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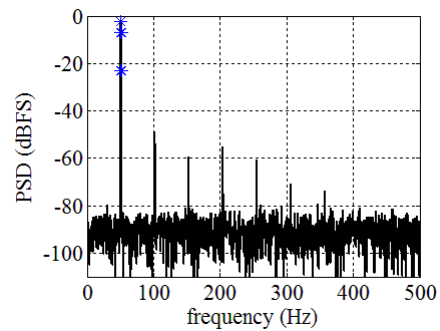


Total Harmonic Distortion

- total harmonic distortion

$$THD = \frac{\text{total distortion power}}{\text{signal power}}$$

- by convention, total distortion power consists of 2nd through 7th harmonic



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Aliasing

- harmonics can appear at “arbitrary” frequencies due to aliasing

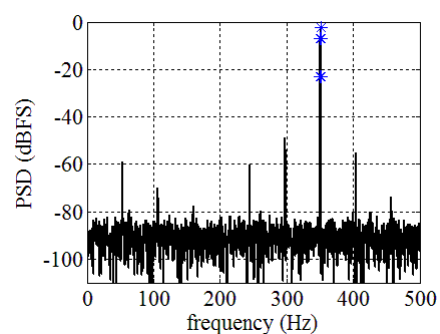
- here:

$$f_{\text{sig}} = 351 \text{ Hz}, f_s = 1 \text{ kHz}$$

$$f_2 = 2f_{\text{sig}} \rightarrow 298 \text{ Hz}$$

$$f_3 = 3f_{\text{sig}} \rightarrow 53 \text{ Hz}$$

$$f_4 = 4f_{\text{sig}} \rightarrow 404 \text{ Hz}$$

$$\vdots$$


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